ELCT 363: Introduction to Microelectronics **Homework #1**
University of South Carolina/ Spring 2022 **Due by: 01/26/2022**
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Student's Name**: Jacob Vaught** **Total Points = 16**

PROBLEM No 1: (4+2 =6 point)

1.A. Identify metals, semiconductors, and insulators among the materials given below **(0.5 point each ➔ total 4 points)**:

Ag	Metal	Au	Metal
Ge	Semiconductor	SiO_2	Insulator
Si	Semiconductor	GaAs	Semiconductor
Ni	Metal	Al_2O_3	Insulator

1.B. Fill in the Blanks **(0.5 point each ➔ total 2 points)**

i. Conductivity of a material depends on: **Free electron concentration__**and

_Mobility of Electrons___________.

ii. At very low temperature (~0 Kelvin) semiconductor behaves as **_an insulator_______.**

iii. **_photons and high voltage_________**can break the bonds in semiconductor crystals and create free electrons.

iv. Mobility and Sheet resistance in semiconductor are measured by

_Voltage and Current_____________??(vand der pauw and hall methods)??

Problem No. 2: (2 points)

A device is powered by a 75 V supply with 0.15 A of current. If the output power of the device is 4.25 W, how much power is wasted in heat?

$$P = V * I$$

Total power consumption: $75V * 0.15A = 11.25W$

Total power wasted: $11.25W - 4.25W = 7W$

Student's Initials: J. C. V.

Problem No. 3 (2 points)

What is the force on an electron in an electric field of 2200 N/C? (**2 points**)

$$\boldsymbol{F = q * E, where\ q\ is\ the\ charge, and\ E\ is\ the\ Electric\ field}$$

$$\boldsymbol{F = 2200\ N/C * 1.6 * 10^{-19} C = 3.52 * 10^{-16} N}$$

Problem No. 4 (2points)

Calculate conductivity (due to free electrons) of an intrinsic (pure) GaAs semiconductor sample at 300K. Assume that free electron concentration is 2.5x 10^6 cm^{-3} and electron mobility is 8200 cm^2/V-s

$$\boldsymbol{\sigma = q * n * \mu, n\ is\ free\ electron\ concentration, and\ \mu\ is\ mobility\ of\ electrons}$$

$$\boldsymbol{\sigma = 1.6 * 10^{-19} * 2.5 * 10^6 * 8200 = 3.28 * 10^{-9} seimen/cm = 328 * 10^{-9} seimen/m}$$

Problem No. 5 (2+1+0.5+0.5 = 4 points)

A silicon slab of 1 cm length with an electron concentration of 3.5x10^{16} cm^{-3} and thickness of 0.07 cm and a cross-section area of 0.19 cm^2 has an electron mobility of 1100 cm^2 /V-sec. Calculate resistivity, sheet resistance, conductivity, and resistance.

$$\text{Conductivity} = \sigma = q * n * \mu$$

$$\sigma = 1.6 * 10^{-19} C * 3.5 * 10^{16} cm^{-3} * \frac{1100 cm^2}{V * sec} = 6.16\ seimen/cm = 616\ seimen/m$$

$$\text{Resistivity} = \rho = \frac{1}{\sigma}$$

$$\rho = \frac{1}{616} = 0.001623\ Ohm * m$$

$$\text{Resistance} = R = \rho * \frac{L}{A}$$

$$R = 0.001623\ Ohm * m * \frac{0.01\ m}{0.000019 m^2} = 0.8542\ Ohm$$

$$R_{sh} = \frac{\rho}{t}$$

$$R_{sh} = \frac{0.001623\ Ohm * m}{0.0007m} = 2.31857$$